Lec: Flight overbooking +
Airbnb Listings

Agenda:

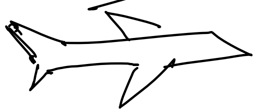
1. Flight overbooking problem - Simple probabilistic modelling

2. Airbnb Listings - Analytics case study using Tableau.

Flight overbooking Problem : (Passenger Bumping)

↳ Denial of service

100 seats (110 bookings)



9 miss ✗

101 show up

100 people ✓✓

1

→ Free meal ✓✓

Alternate flight ✓✓

Hotel stay ✓✓

DGCA

↳ (Regulatory body)

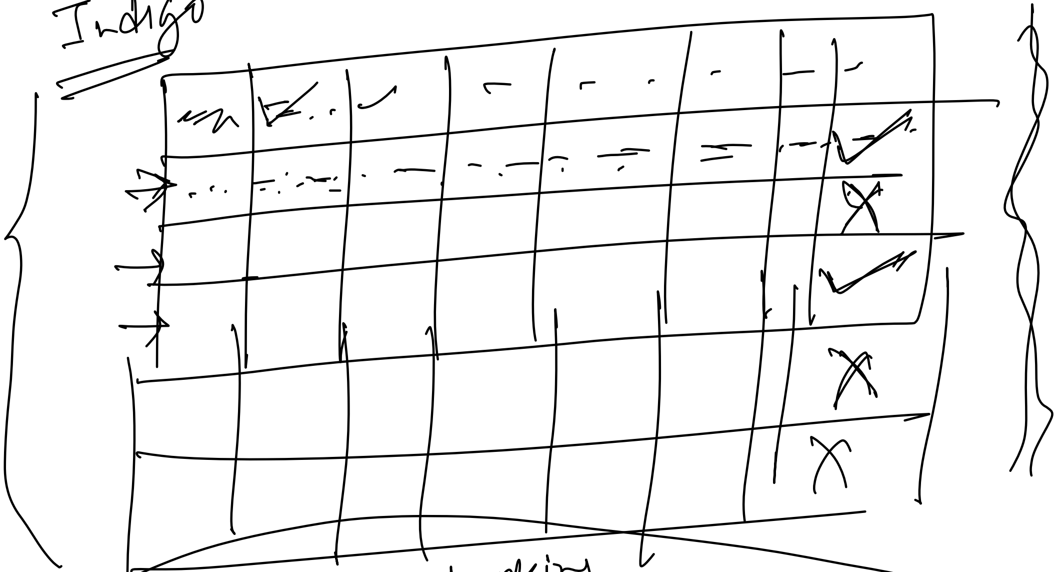
↳ Penalty

Problem stmt:

Figure out the optimal number of extra tickets that can be sold to maximize revenue, without incurring too much penalty.

Simple Probabilistic model

Indigo



90% show up

100 passengers * 90%

= 90 people show up.

extra booking

$$(100+x) * 90\% \leq 100$$

$$0.9x + 90 \leq 100$$

$$\Rightarrow x \leq \frac{10}{0.9} \approx \lfloor 11.11 \rfloor = 11 \text{ seats}$$

floor function

$$100 + x = 100 + \underbrace{11}_{\text{extra seats}} = 111$$

$\boxed{\text{This model may sometime fail}}$

Improved Model :

~~100 total seats~~ (110 booked)

102 people show up

↳ 2 person bumped

Seat price = Rs 5K

Penalty = Rs 10K → 2 person

↳ 20K rupees.

Revenue

110 * 5K

↳ 5.5L

Case I : 100 seats in flight

110 tickets sold → 10 overbookings.

100 passenger arrived

↳ No bumping

Revenue generated = 110 * 5K

= 5,50,000 rupees.

Penalty = 0

Case II : $\dfrac{\text{100 seats} \rightarrow \text{Flight}}{\text{110 tickets booked}}$

↳ 101 passenger show up

↳ 1 person got bumped.

↳ 10K penalty

↳ Revenue : 5,50,000

↳ Profit : 5,40,000

90% chance show up
10% " NO show up

110

101   110   100

X   X   ✓

$P(\text{showing up}) = 0.9$

$P(\text{Not showing}) = 0.1$

101 people show up = 1 extra person show up

$$P(X=101) = {}^{110}C_{101} * \underbrace{(0.9 * 0.9 * 0.9 * \ldots * 0.9)}_{101 \text{ times}}$$

$$* \underbrace{(0.1 * 0.1 * \ldots * 0.1)}_{9 \text{ times}}$$

$$= {}^{110}C_{101} * (0.9)^{101} * (0.1)^{9}$$

≈ 0.11

$\approx 11\%$

Expected (Avg) penalty $= 10000 * 11\%$

$= 1100$

$$\text{Wt. Avg penalty} = \frac{\Sigma(\text{value} * \text{weights})}{\Sigma \text{weights}}$$

$$= \frac{(10000) * 11\% + 0 * 89\%}{(0.11 + 0.89)}$$

$$= \frac{10K * 11\% + 0}{1} = 1100$$

$P(X=102)$, $P(X=103)$, $P(X=104)$, ..., $P(X=110)$

X (People Showed Up)	Extra Passengers	Penalty = ₹10k × Extra	Probability (%)	Expected Penalty (₹)
101	1	₹10,000	11.0%	₹1,100
102	2	₹20,000	8.8%	₹1,760
103	3	₹30,000	6.1%	₹1,830
104	4	₹40,000	3.3%	₹1,320
105	5	₹50,000	1.5%	₹750
106	6	₹60,000	0.7%	₹420
107	7	₹70,000	0.3%	₹210
108	8	₹80,000	0.15%	₹120
109	9	₹90,000	0.05%	₹45
110	10	₹1,00,000	0.01%	₹10

110 booking

Total Expected penalty

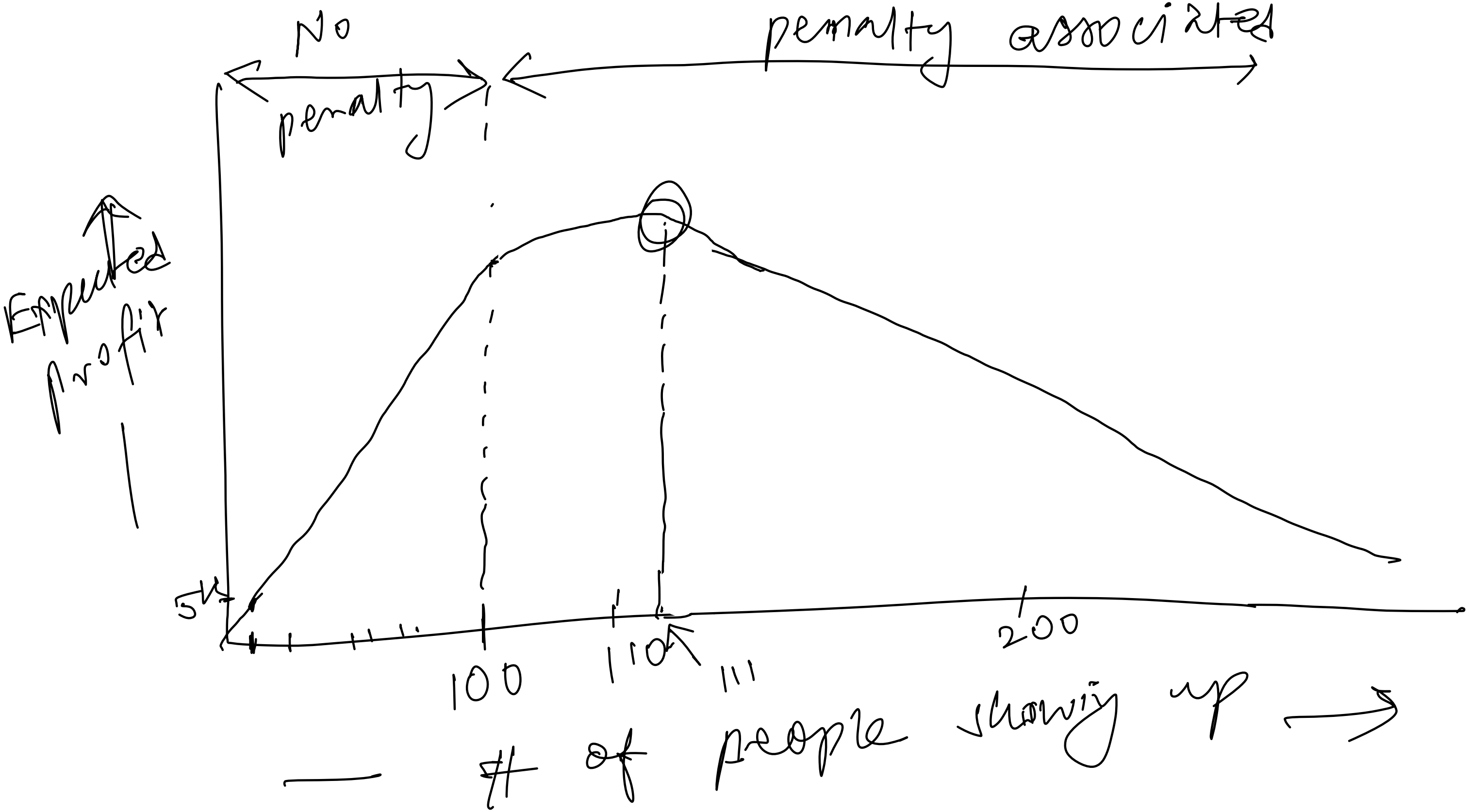
$= 1100 + 1760 + \ldots + 10$

$= 7565$ rupees

Expected profit = Revenue − Total Expected penalty

$= 5.5L - 7565$

$= 542435$ rupees.


No penalty
penalty associated
Expected profit
5k
100
110
111
200
# of people showing up

Extra Seats Sold	Total Tickets Sold	Expected Net Revenue (₹)
0	100	500,000.00
1	101	504,999.76
2	102	509,997.13
3	103	514,982.43
4	104	519,926.68
5	105	524,765.24
6	106	529,383.97
7	107	533,617.97
8	108	537,269.40
9	109	540,141.64
10	110	542,078.85
11	111	542,996.88
12	112	542,895.73
13	113	541,851.84
14	114	539,995.73
15	115	537,484.06
16	116	534,474.15
17	117	531,106.14
18	118	527,493.77
19	119	523,722.57
20	120	519,852.74
21	121	515,924.21
22	122	511,962.14
23	123	507,981.63
24	124	503,991.33
25	125	499,996.02
26	126	495,998.22
27	127	491,999.22
28	128	487,999.67
29	129	483,999.86
30	130	479,999.94

Airbnb - Tableau

Higher Property images ⟷ Higher property bookings.

??

After analysis (Tableau)

5 images → ↑↑↑↑

Recommendation :-

1. minimum 5

2. optimal 6 - 8